Word Skipping in L1 and L2 English Readers: A Multifaceted Analysis of Parafoveal Processing

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Introduction: word skipping

The quick brown fox ~~jumps~~ over the lazy dog.

focus

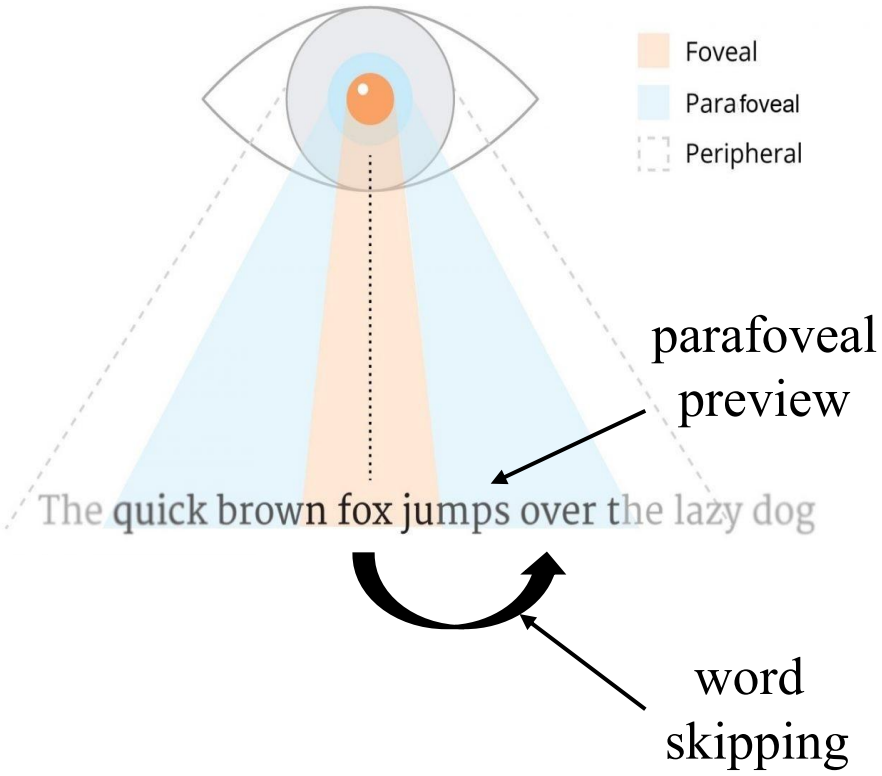
Word skipping occurs when readers' eyes move past a word without fixating it, a natural but unconscious process captured through eye tracking. (Rayner, 1998).

Word skipping reflects both **visual word recognition efficiency** and **linguistic prediction.**
(Rayner, 1998; Reichle et al., 1998; Staub, 2015).

The current study investigates the **mechanisms** underlying how L2 English readers differ from L1 readers in word skipping during their initial pass through the text, as known as ***first-pass skipping***.

Introduction: parafoveal processing

Word skipping is based on **parafoveal processing**, which enables readers to extract orthographic, phonological, and semantic information from the **parafoveal preview** (1–5° visual angle from central fovea).
(Liversedge & Findlay, 2000)



Competing theories of skipping

◦ *Autonomous Oculomotor Scanning Strategy*

Skipping is largely driven by **low-level visual control** and text layout (McConkie & Yang, 2003).

◦ *Language Control View*

Skipping is guided by **ongoing linguistic processing,** since words have been processed sufficiently in parafoveal vision before the saccade is initiated (Brysbaert et al., 2005).

◦ *Extended Optimal Viewing Position* (EOVP) *model*

Skipping reflects an **educated guess** based on **word length** and estimated comprehension cost (Brysbaert & Vitu, 1998).

Major factors affecting skipping

Oculomotor Factors

1. Word length

shorter → higher skipping

(Schotter & Dillon, 2025)

2. Launch site distance of the saccade, an index of parafoveal preview quality, i.e., the distance from the current fixation (word *n*) to the beginning of the next word (*n*+1)

shorter → higher skipping

(McDonald & Shillcock, 2003; Brysbaert et al., 2005; Milligan, 2025)

Linguistic Factors

1. Word frequency

higher → higher skipping

(Staub et al., 2010)

2. Predictability

higher → higher skipping

(Brysbaert et al., 2005; Staub et al., 2010)

Previous L2 skipping research

◦ Research on Multilingual Eye-Movement Corpus (MECO)

L2 skipping is shaped not only by **word length** and **frequency** but also by L1 reading habits. (Esteve et al., 2024)

◦ The *Reduced Capacity for Parafoveal Processing* (ReCaPP) hypothesis

L2 readers show similarity to L1 readers in terms of **predictability**, but they possess a **reduced capacity** for the parafoveal preview. (Fernandez & Allen, 2025)

◦ Previous L2 studies

The effects of word length, frequency, and predictability ✓

Interaction between these effects and **parafoveal preview quality** ✗

The **relative weighting** of oculomotor versus linguistic factors ✗

Research questions

1. How do oculomotor and linguistic factors influence ***first-pass skipping*** during L1 and L2 English reading?

2. How does the ***first-pass skipping*** differ between L1 and L2 English readers?

Defining oculomotor factors

Word length

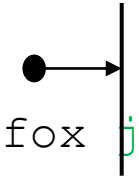
# of letter spaces (+ punctuation)

Launch site distance

# of letter spaces between the fixation and subsequent words

→ key index of **parafoveal preview quality**

Launch site distance for “jumps”
= 3 letter spaces



The quick brown fox jumps over the lazy dog.

Defining linguistic factors

Word frequency

$SUBTLEX_{US}$ word frequency (Brysbaert & New, 2009).

Word predictability

GPT Surprisal (negative log-predictability) from the GPT-2 model (Radford et al., 2019)

→ expectation-based metric that captures the predictability of each word in context (Berzak & Levy, 2023)

Methodology and data

Method:

Eye-tracking during reading conducted at University of Texas at Arlington Eye-Tracking Lab

Materials:

6 English reading passages

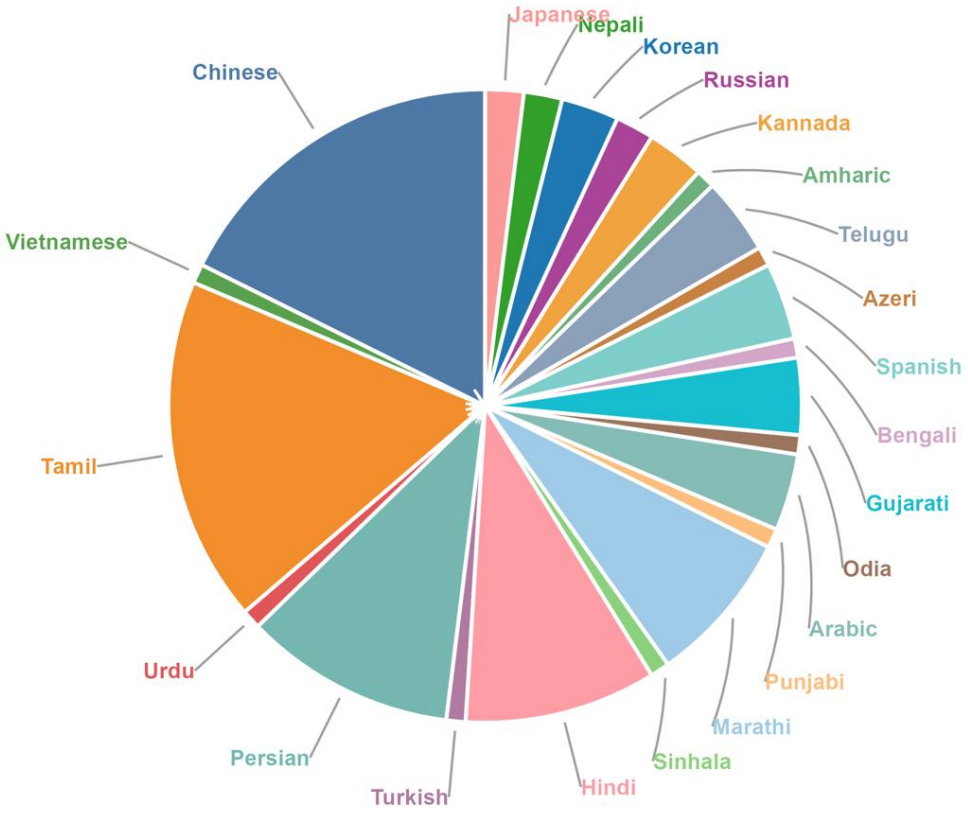
(3 sections each)

Total participants: 190

L1 English participants: 88

L2 English participants: 102

23 different L1 backgrounds in the L2 English group: Chinese (18), Tamil (18), Persian (11), Hindi (10), Marathi (8), Gujarati (4), Spanish (4), Arabic (4), Korean (3), Telugu (4), Kannada (3), Azeri (1), Japanese (2), Nepali (2), Russian (2), Amharic (1), Bengali (1), Odia (1), Punjabi (1), Sinhala (1), Turkish (1), Urdu (1), and Vietnamese (1).



Analytical approach: logistic linear mixed effects model

GLMM:

Word skipping (binary outcome) ~

Word length
Launch site distance

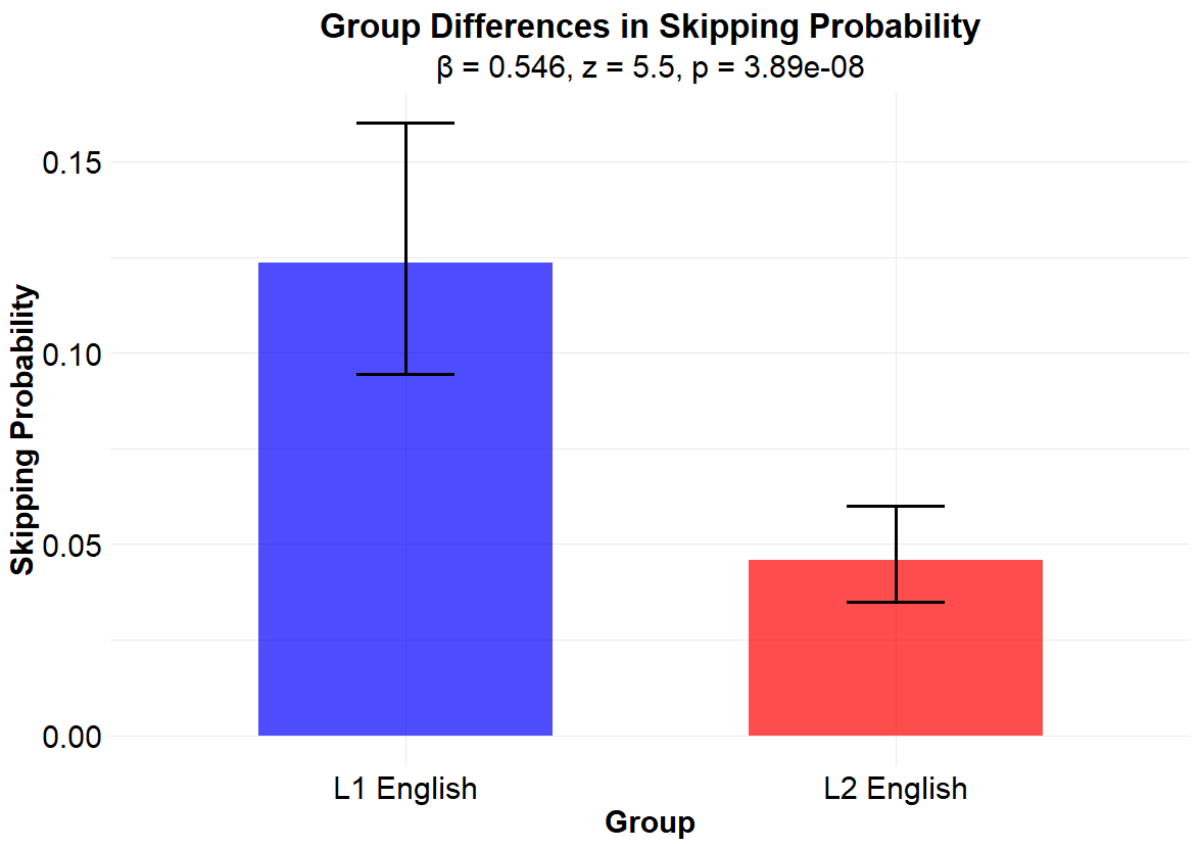
} Oculomotor factors

Word frequency
Word predictability (GPT surprisal)

} Linguistic factors

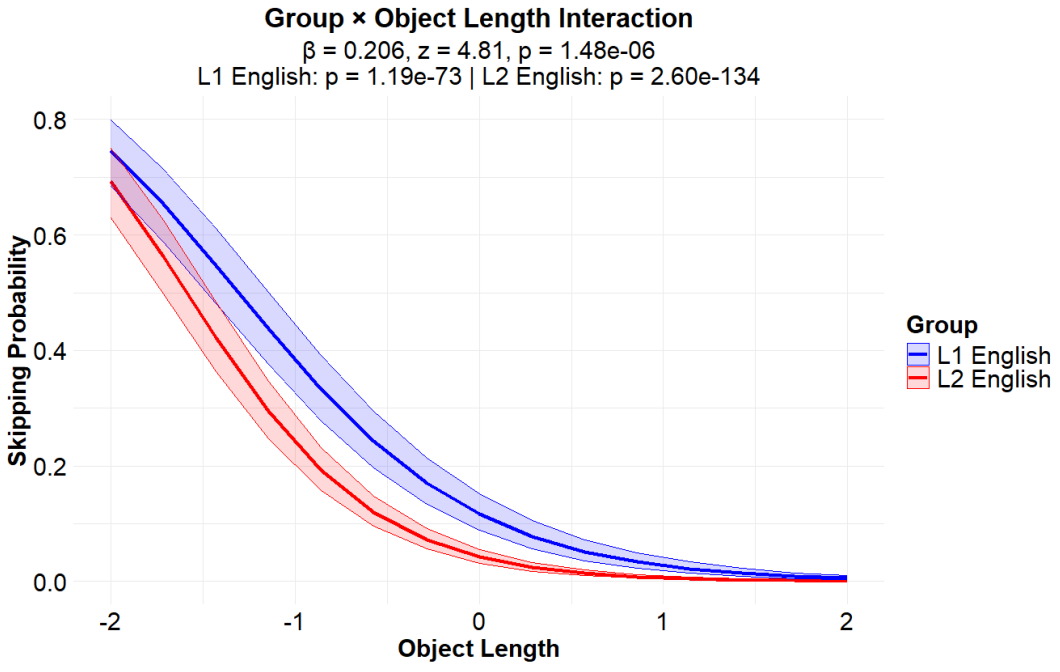
Group: L1 English readers vs. L2 English readers.

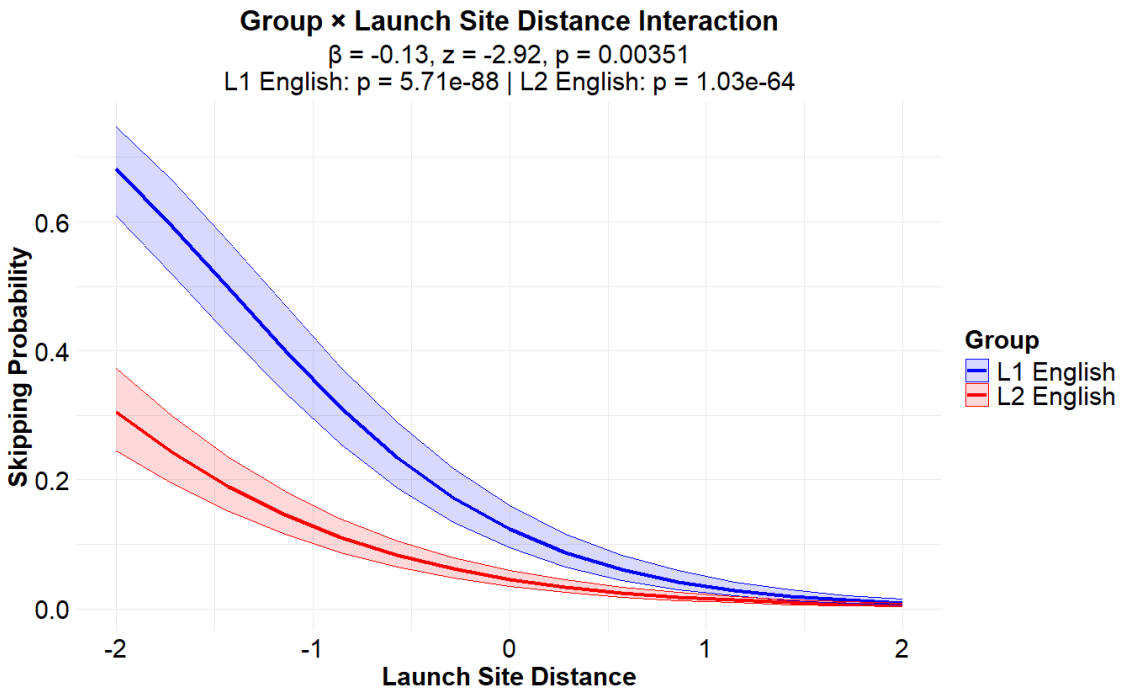
Key findings (groups)



L1 readers skip words more often than L2 readers (β=0.55, p<.001).

Key findings (oculomotor factors)





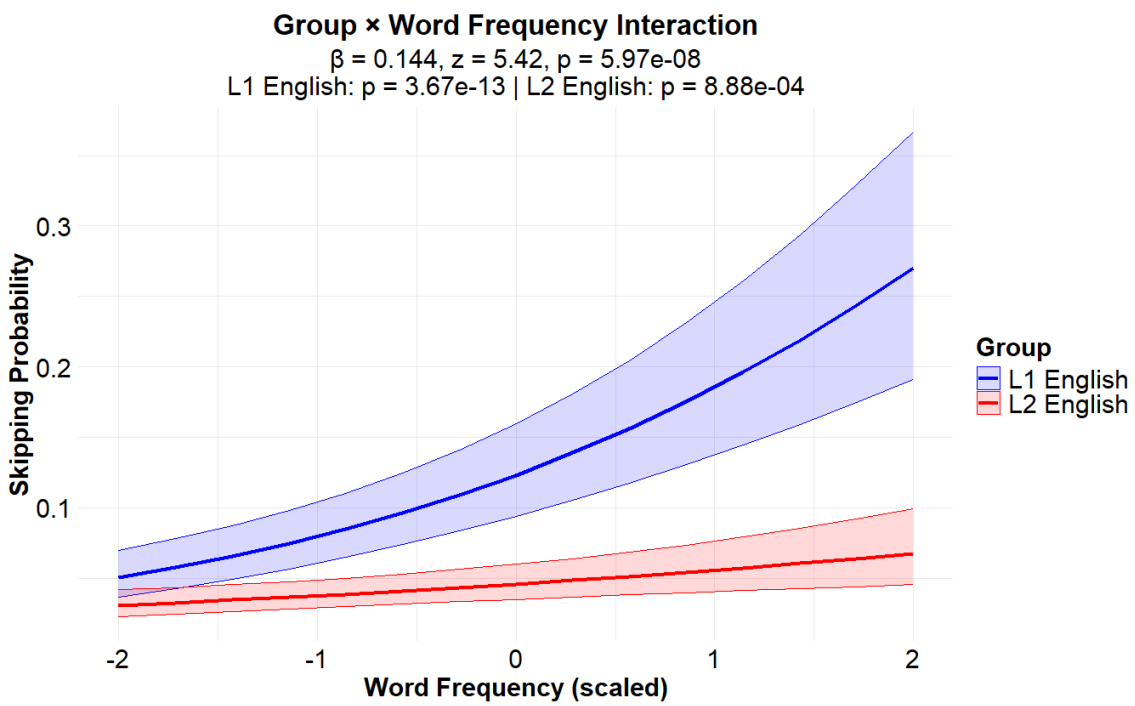
L2 readers show a higher skipping rate (β=-1.76, p<.001)

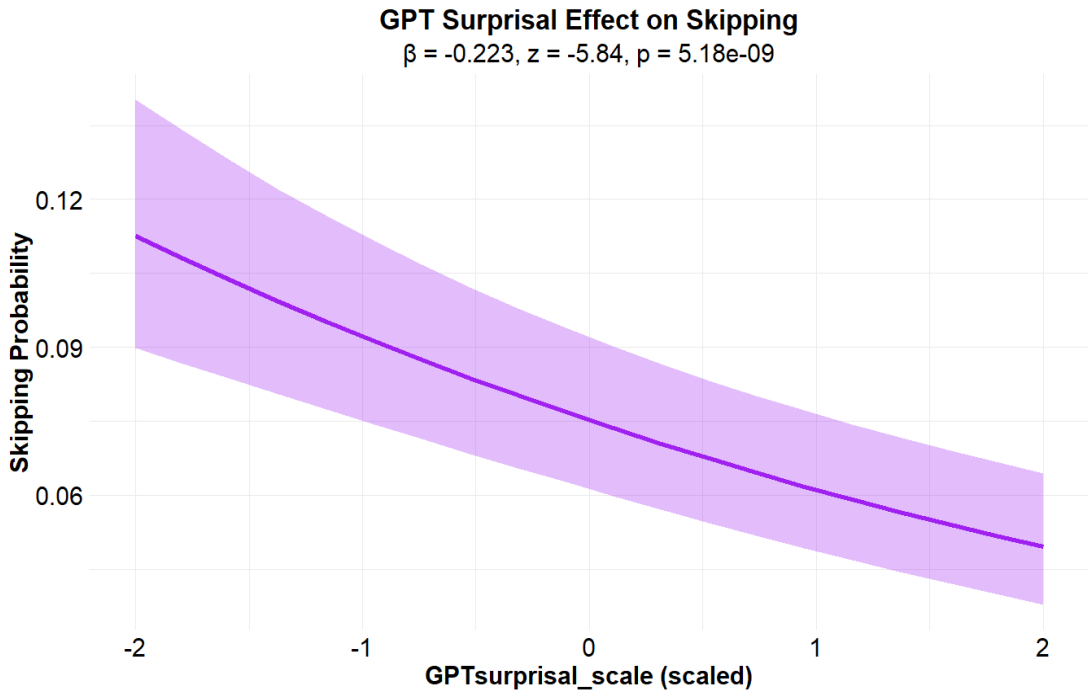
Shorter words have higher skipping rates, increasing word length (group × length: β=0.21, p<.001)

L2 readers show a smaller effect of launch site distance (group × launch site: β=-0.13, p<.01).

Shorter launch site distances boosts skipping (β=-1.22, p<.001)

Key findings (linguistic factors)





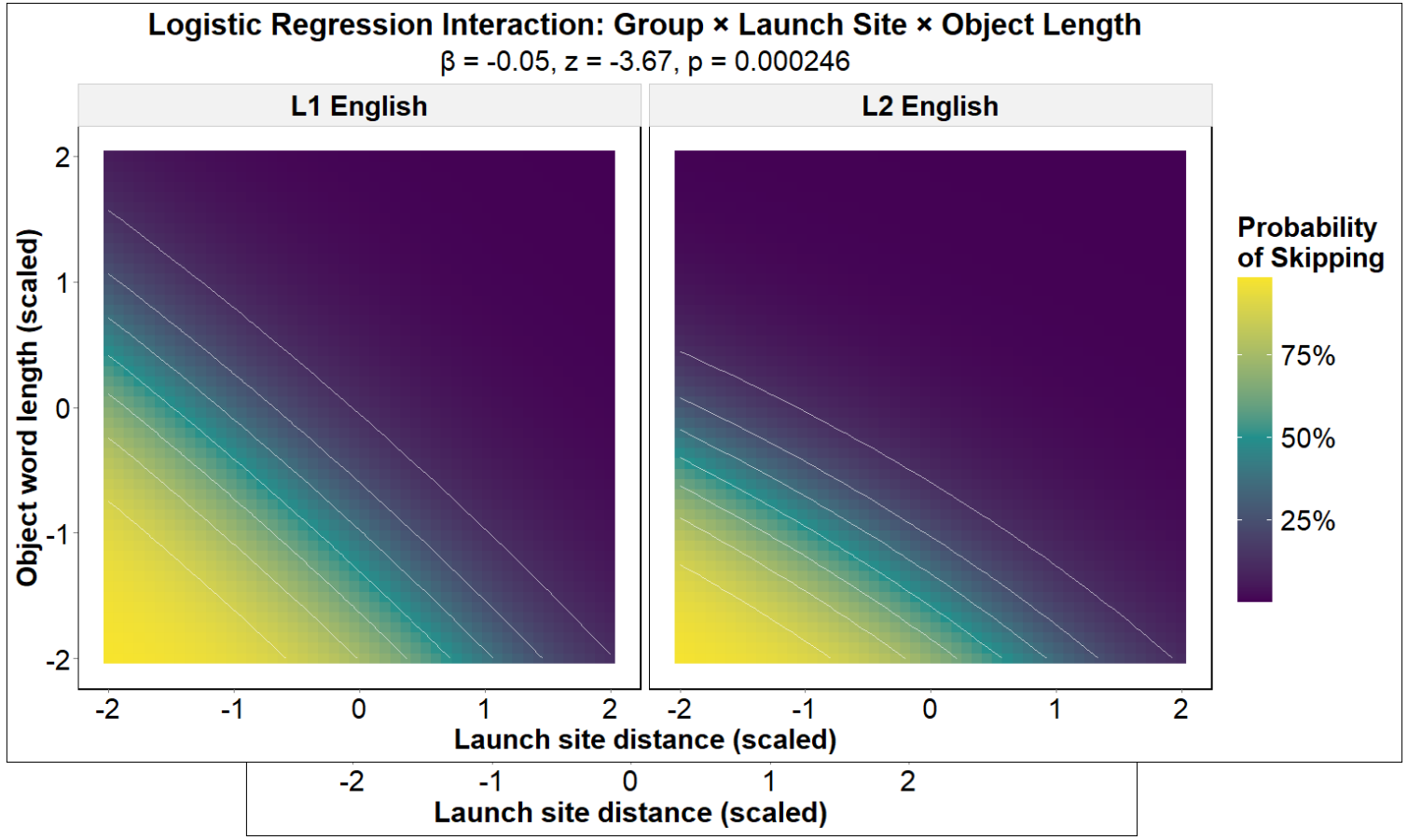
Higher word frequency yields a larger increase in skipping for L1 readers (group × frequency: β=0.14, p<.001).

High-frequency words are skipped more often (β=0.33, p<.001).

GPT surprisal shows no reliable skipping interaction, p<.001).

GPT surprisal increases skipping (β=-0.22,

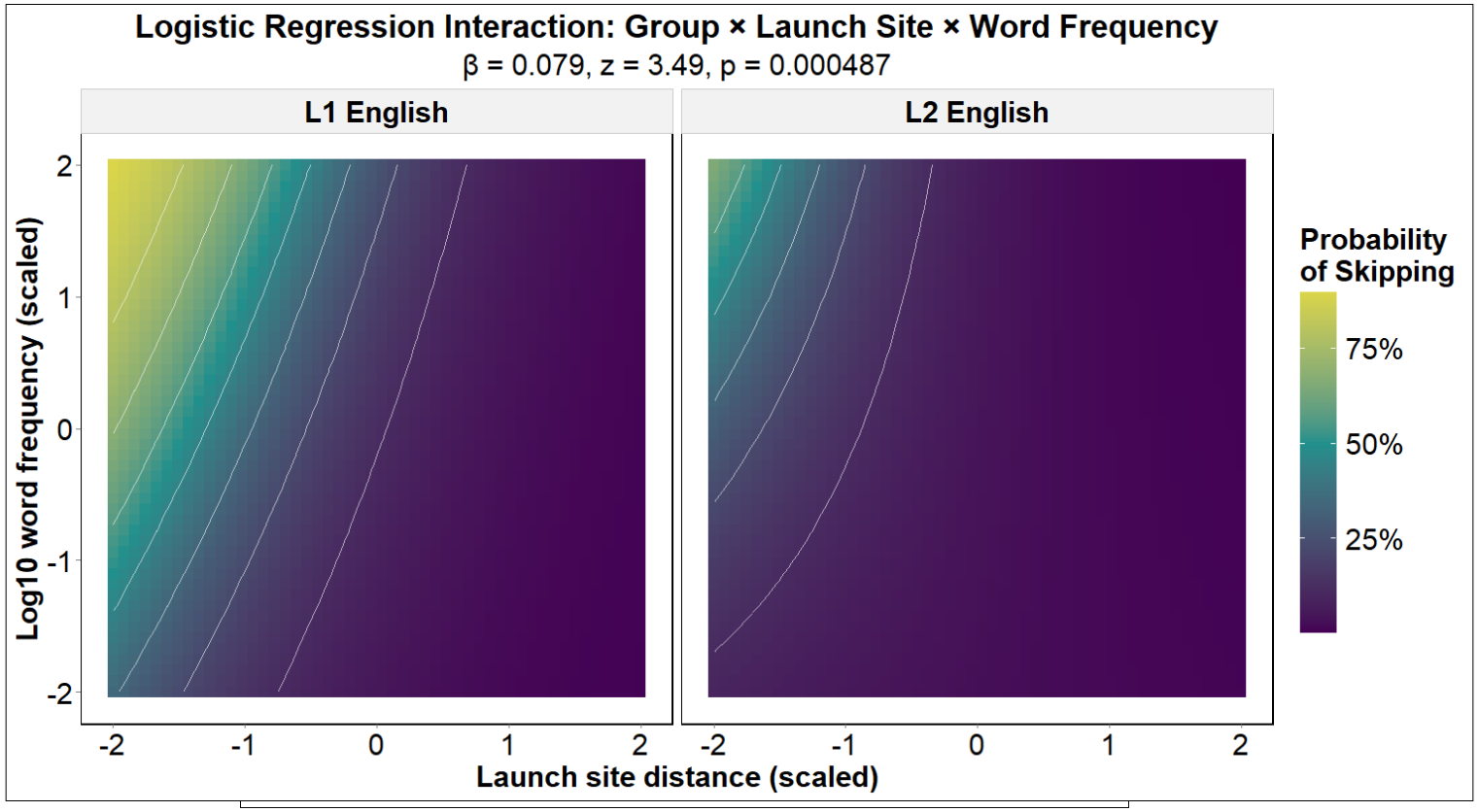
Key findings (parafoveal preview x word length)



L2 readers show a sharper change in skipping rate as both word length and launch site decrease (β=-0.05, p<.001)

Higher skipping rates for shorter words and closer launch sites

Key findings (parafoveal preview x word frequency)



Higher skipping rates for more frequent words at shorter launch site distances.

Higher skipping rates for more frequent words for L2 readers.

Main takeaways

- The results clearly indicate that parafoveal preview quality can determine whether shorter words (β=0.12, p<.001) and higher frequency (β=-0.19, p<.001) affect skipping.

- L2 readers show **higher** sensitivity to **oculomotor** factors such as word length, but **lower** sensitivity to **linguistic** factors such as word frequency compared to L1 readers.

- The study supports a hybrid theoretical account: oculomotor and linguistic factors jointly guide skipping, with L1 & L2 readers showing different reliance on these factors.

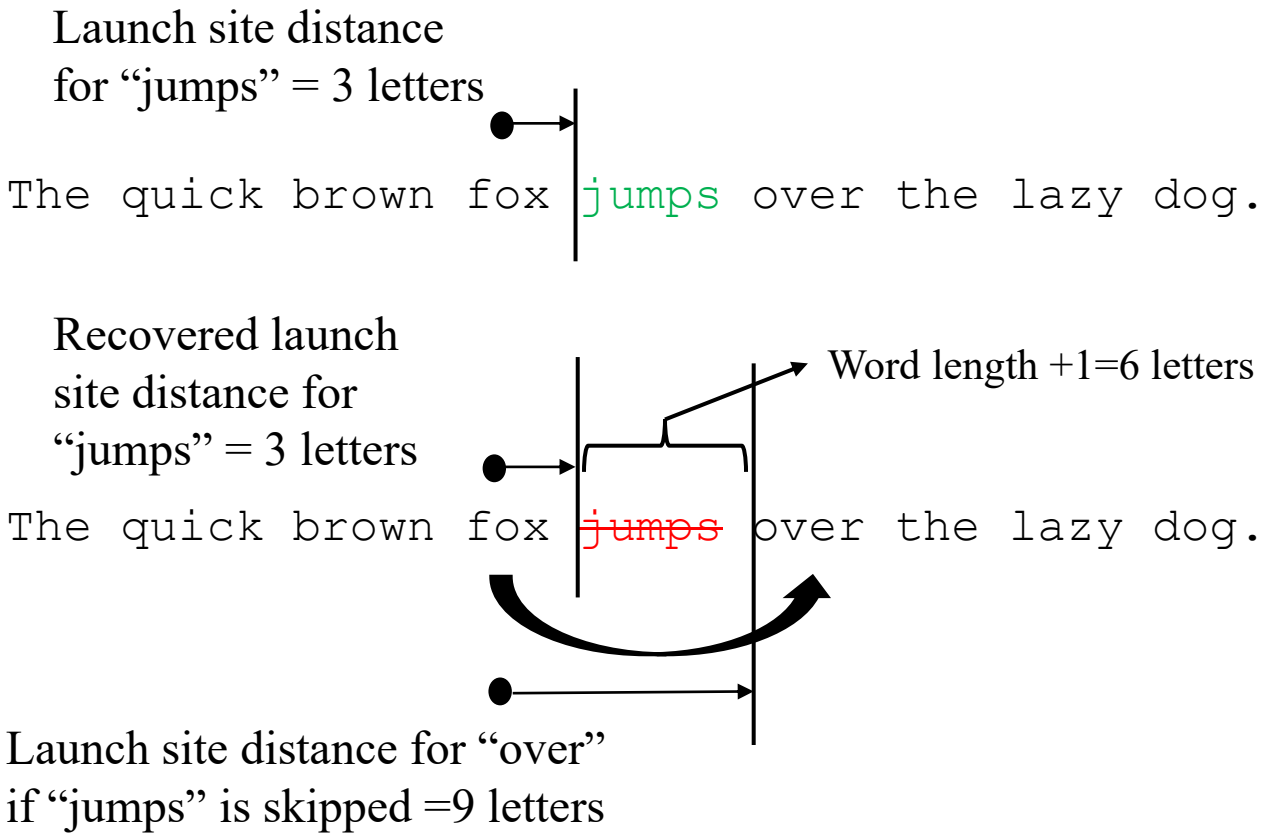
Acknowledgement

Many thanks to Dr. Jeffrey Witzel and Dr. Naoko Witzel as well as all of the faculty, students and friends from the Department of Linguistics & TESOL at UT Arlington.

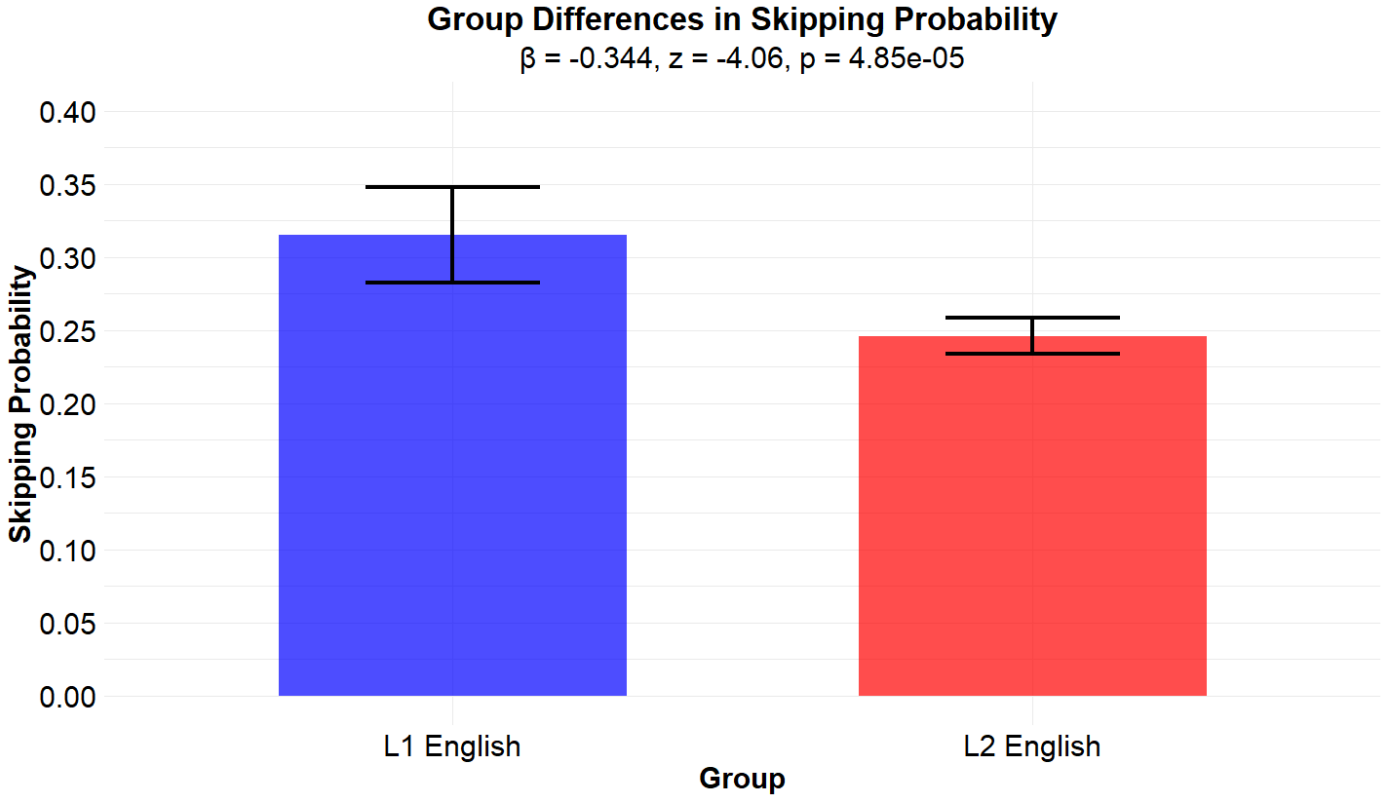
Thank you very much for listening!

Appendix: measure of launch site distance

The English text for eye tracking is presented in 18-point monospaced (`Courier New`) font with double-line spacing on a 19-inch CRT monitor. The monitor screen is located ~60 cm from the participant's eyes. At this viewing distance, three letters equal one degree of visual angle.



Appendix: overall group difference



L1 readers skip words more often than L2 readers overall (β=-0.34, p<.001).